

7.5 Indeterminate Forms and L'Hôpital's Rule

Expressions such as " $\frac{0}{0}$ " and " $\frac{\infty}{\infty}$ " look something like ordinary numbers. We say that they have the *form* of a number. But values cannot be assigned to them in a way that is consistent with the usual rules to add and multiply numbers. We call these expressions "indeterminate forms." Although they are not numbers, these indeterminate forms play a useful role in summarizing the limiting behavior of a function.

John (Johann) Bernoulli discovered a rule using derivatives to calculate limits of fractions whose numerators and denominators both approach zero or ∞ . The rule is known today as **L'Hôpital's Rule**.

① Indeterminate Form $\frac{0}{0}$

THEOREM 5 — L'Hôpital's Rule

Suppose that $f(a) = g(a) = 0$, that f and g are differentiable on an open interval I containing a , and that $g'(x) \neq 0$ on I if $x \neq a$. Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)},$$

assuming that the limit on the right side of this equation exists.

Example 1 Find the following limits.

(a) $\lim_{x \rightarrow 0} \frac{3x - \sin x}{x}$

(b) $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$

(c) $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1 - \frac{x}{2}}{x^2}$

(d) $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$

Note 1 To find $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ by L'Hôpital's Rule, we continue to differentiate f and g , so long as we still get the form $\frac{0}{0}$ at $x = a$. But as soon as one or the other of these derivatives is different from zero at $x = a$ we stop differentiating. L'Hôpital's Rule does not apply when either the numerator or denominator has a finite nonzero limit.

Example 2 Find the limit $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x + x^2}$.

Note 2 L'Hôpital's Rule can be applied to one-side limits as well.

Example 3 Find the limit $\lim_{x \rightarrow 0^+} \frac{\sin x}{x^2}$.

Example 4 Find the limit $\lim_{x \rightarrow 0^-} \frac{\sin x}{x^2 + x}$.

② Indeterminate Forms $\frac{\infty}{\infty}, \infty \cdot 0, \infty - \infty$

More advanced treatments of calculus prove that L'Hôpital's Rule applies to the indeterminate form $\frac{\infty}{\infty}$, as well as to $\frac{0}{0}$. If $f(x) \rightarrow \pm\infty$ and $g(x) \rightarrow \pm\infty$ as $x \rightarrow a$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)},$$

provided the limit on the right exists.

Example 5 Find the following limits.

(a) $\lim_{x \rightarrow \infty} \frac{\ln x}{2\sqrt{x}};$

(b) $\lim_{x \rightarrow \infty} \frac{e^x}{x^2}.$

Next, we turn our attention to the indeterminate forms $\infty \cdot 0$ and $\infty - \infty$. Sometimes

these forms can be handled by using algebra to convert them to a $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form.

Example 6 Find the following limits.

(a) $\lim_{x \rightarrow \infty} x \sin \frac{1}{x}$; (b) $\lim_{x \rightarrow 0^+} \sqrt{x} \ln x$.

Example 7 Find the limit $\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right)$.

Example 8 Find the limit $\lim_{x \rightarrow \infty} \left(x - \sqrt{x^2 + x} \right)$.

Example 9 Find the limit $\lim_{x \rightarrow 0} \frac{\tan x (x - \sin x)}{x^4}$.

Example 10 For what values of a and b is

$$\lim_{x \rightarrow 0} \left(\frac{\tan 2x}{x^3} + \frac{a}{x^2} + \frac{\sin bx}{x} \right) = 0.$$